

EC824B

Introduction

Half Course outline

- High-dimensional econometrics
- Some buzzwords: machine learning, AI, Big Data

$$Y = h(X, Z, U)$$

Usually in economics we are interested in a **low-dimensional object**, sometimes in a **high-dimensional setting**

As we get better estimators, more/better data we can:

1. Admit more general functions $h(\cdot, \cdot, \cdot)$
2. More robustly control Z
3. Control for more general unobserved heterogeneity, U

without compromising a clean interpretation of X 's impact on Y

Bet on sparsity

Use a procedure that does well in sparse problems, since no procedure does well in dense problems.

- Hastie, Tibshirani, and Wainwright

- Economic problems are usually sparse!
 1. Out of potentially hundreds (or thousands) of variables, we are usually only interested in one or two (others are “controls” Z)
 2. In panel data, sometimes interested in inference on unobserved heterogeneity, but usually we just want to control for it
 3. Even when we have complicated functions of X , we are usually still interested in some aggregate, e.g. ATE, or price elasticity
- If we don't allow a sparse solution, we may have no hope of meaningful inference

Sparsity 1

X is very high-dimensional, e.g. $X \in \mathbb{R}^{N \times p}$ with $p \gg N$, but the model is otherwise simplified:

$$Y = X'\delta + U$$

For example Best subset (infeasible), LASSO, and Ridge regression to make estimation of δ and prediction of Y feasible

- Sparsity here is bounding the set of p potential variables that are non-zero, or bounding their contribution through “shrinkage” on δ

Sparsity 2

If X has granular support (e.g. $X \in \mathbb{R}$) we need to simplify the function,

$$h(X, Z, U) = X\beta + g(Z) + U$$

For example [Belloni, Chernozhukov and Hansen (2014)]

Sparsity 3

If X is “low-dimensional” (e.g. $X \in \{0, 1\}$) we can maintain a generic function for X , but lower dimension for Z :

$$h(X, Z, U) = h(X, U) + Z'\delta$$

For example [Athey et al. (2021); Fernández-Val, Freeman and Weidner (2021)] , which can control for very generic unobserved heterogeneity U too

Sparsity 4

For panel data models, unobserved heterogeneity U can be generalised, but sparsity must remain!

- Sparsest:

$$U_{it} = \alpha_i + \varepsilon_{it} \quad \text{or} \quad U_{it} = \gamma_t + \varepsilon_{it}$$

- Sparse:

$$U_{it} = \alpha_i + \gamma_t + \varepsilon_{it}$$

- Sparse(ish): Usually requires large N and T [Hotelling (1933); Bai (2009)]

$$U_{it} = \alpha_i' \gamma_t + \varepsilon_{it}$$

- Also sparse(ish): Usually requires large N and T [Fernández-Val, Freeman and Weidner (2021); Freeman and Weidner (2023); Griebel and Harbrecht (2019)]

$$U_{it} = g(\alpha_i, \gamma_t) + \varepsilon_{it}$$

Bias-variance trade-off

One very important concept in high-dimensional settings is how to trade-off bias and variance

- We see in the sequel that often, to have low bias, we must need a very complicated model (more parameters)
- This often induces higher variance (more to estimate, lower degrees of freedom)
- If we give up some bias (reduce parameters), we can reduce variance (less to estimate, higher dof)
- When to stop? Trade-off!

In the above high-level examples we saw that in general greater sparsity bought us back some variance, but at the cost of being less general (more bias).

- We will see that even our beloved OLS estimator can be improved with a bit of sparsity

Course Assessment

Take-home exam (100%)

“
When you're fundraising, it's AI.
When you're hiring, it's ML.
When you're implementing, it's logistic regression.”

References I

- Athey, S., M. Bayati, N. Doudchenko, G. Imbens, and K. Khosravi (2021). Matrix completion methods for causal panel data models. *Journal of the American Statistical Association* 116(536), 1716–1730.
- Bai, J. (2009). Panel data models with interactive fixed effects. *Econometrica* 77(4), 1229–1279.
- Belloni, A., V. Chernozhukov, and C. Hansen (2014). Inference on treatment effects after selection among high-dimensional controls. *Review of Economic Studies* 81(2), 608–650.
- Fernández-Val, I., H. Freeman, and M. Weidner (2021). Low-rank approximations of nonseparable panel models. *The Econometrics Journal* 24(2), C40–C77.
- Freeman, H. and M. Weidner (2023). Linear panel regressions with two-way unobserved heterogeneity. *Journal of Econometrics* 237(1), 105498.
- Griebel, M. and H. Harbrecht (2019). Singular value decomposition versus sparse grids: refined complexity estimates. *IMA Journal of Numerical Analysis* 39(4), 1652–1671.

References II

Hotelling, H. (1933). Analysis of a complex of statistical variables into principal components. *Journal of educational psychology* 24(6), 417.